PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORK WALKS

Details	Information
Pentester Name (Cybersecurity Trainee)	Harini Madasu
Program/Batch	B084 – Network Walks
Report Date	12 November 2026
Modules Completed	W2-PM1 – Web Footprinting & Information Gathering; W2-PM5 – Zenmap Network Scanning
Client/Target	1. example.com – Reserved domain used for educational testing
	2. 127.0.0.1 – Own local system (localhost)
Permission / Scope	Testing restricted to the reserved example.com domain and the trainee's own local system. No unauthorized network scanning was performed.
Phases Covered	Phase 1: Reconnaissance & Footprinting; Phase 2: Network Scanning & Analysis

---

1. Liability Disclaimer

The activities documented in this report were performed for authorized educational and training purposes. The footprinting exercise was conducted against example.com, a reserved domain intended for documentation and testing, while the network scanning exercise was restricted to my own local system using the loopback address 127.0.0.1.

No unauthorized scanning of other devices on the college network was performed.

All findings are intended for educational and research purposes only. The tools and techniques discussed in this report must not be used against systems or networks without appropriate authorization. Any misuse of cybersecurity knowledge may result in legal consequences. The trainee is responsible for ensuring that all security testing remains within the permitted scope.

2. Introduction

This report documents the practical cybersecurity activities completed during Week 2 of my training program at Network Walks. The work covered two major areas: web footprinting and reconnaissance (W2-PM1), followed by network scanning and analysis using Zenmap (W2-PM5).

The first activity focused on collecting publicly available information about the example.com domain. It involved examining domain registration details, identifying website technologies, retrieving DNS information, detecting Web Application Firewall (WAF) protection, and performing DNS enumeration.

The second activity involved using Zenmap, the graphical interface for Nmap, to perform network scanning against my own Windows system through the localhost address 127.0.0.1. The exercises included host discovery, TCP port examination, service version detection, and an attempt at operating system detection.

PM1 was performed using Kali Linux running through Oracle VirtualBox, whereas PM5 was performed using Zenmap and Nmap on Windows.

The practical activities were documented using command outputs, scan results, screenshots, and a topology PDF. This report presents the activities performed, observations, potential security implications, and recommendations.

3. Tools Used

The following tools and technologies were used during the practical activities.

Tool / Technology	**Purpose**
Kali Linux	Linux-based cybersecurity testing environment used for PM1
Oracle VirtualBox	Virtualization platform used to run Kali Linux
WHOIS	Retrieves publicly available domain registration information
WhatWeb	Identifies website technologies and web server information
Nslookup	Resolves domain names and retrieves DNS records
Wafw00f	Detects Web Application Firewalls
DNSRecon	Performs DNS enumeration and gathers DNS-related information
Windows	Operating system used for PM5

Zenmap	Graphical interface for Nmap used for network scanning
Nmap	Performs host discovery, port scanning, service detection, and OS detection
Windows Command Prompt	Used to examine IP configuration and MAC addresses

4. Activities Performed

4.1 W2-PM1: Web Footprinting & Reconnaissance

For the first activity, I performed web footprinting and information gathering against the example.com domain using five Kali Linux tools: WHOIS, WhatWeb, Nslookup, Wafw00f, and DNSRecon. Each tool was used to collect a different category of publicly available information about the target.

1. WHOIS – Domain Information Gathering

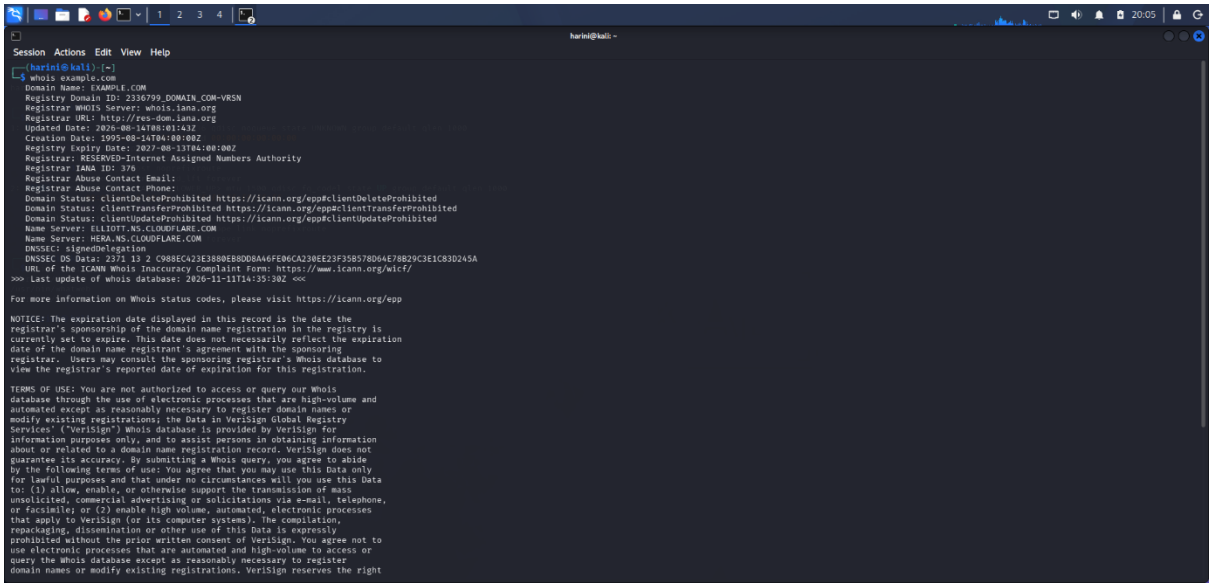
Objective: Retrieve publicly available registration information associated with the target domain.

Command used:

whois example.com

Observation: The command executed successfully and displayed domain registration-related information, including registrar and name server details.

This activity helped me understand how WHOIS can be used during the reconnaissance phase to collect publicly available information about a domain.



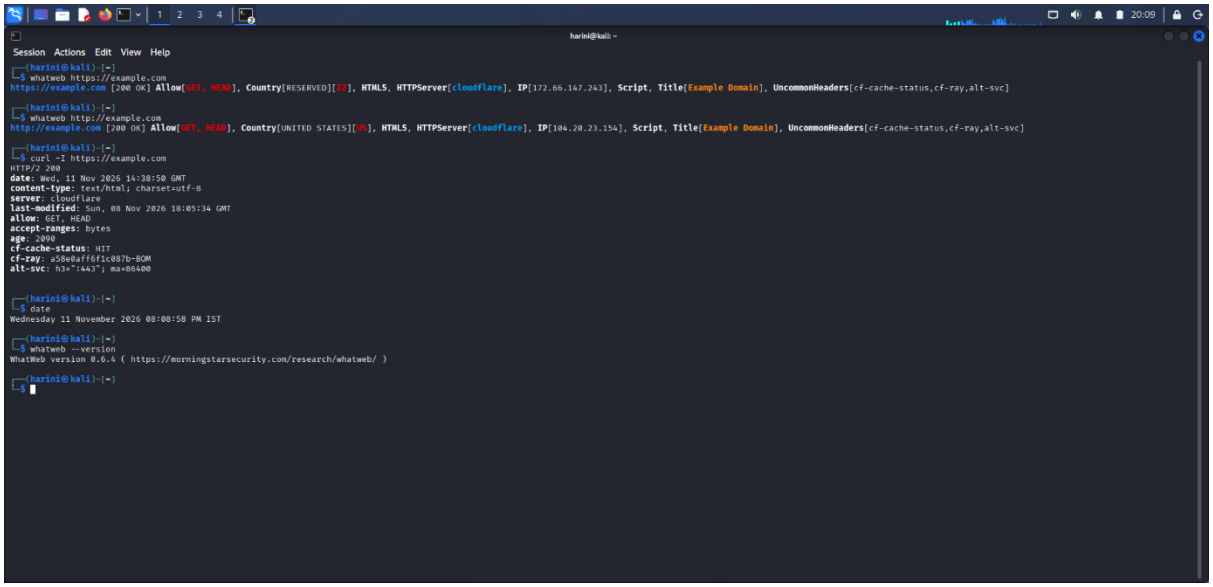
2. WhatWeb – Website Technology Identification

Objective: Identify technologies and web-related information associated with the target website.

Command used: whatweb http://example.com

Observation: The command executed successfully and displayed detected website information. The WhatWeb version used was 0.6.4. An initial HTTPS attempt returned an SSL connection error. Therefore, the successful identification was performed using the HTTP address.

This activity demonstrated how website fingerprinting tools can help identify technologies and gather information useful during security assessments.



3. Nslookup – DNS Information Gathering

Objective: Retrieve DNS information associated with the target domain.

Commands used:

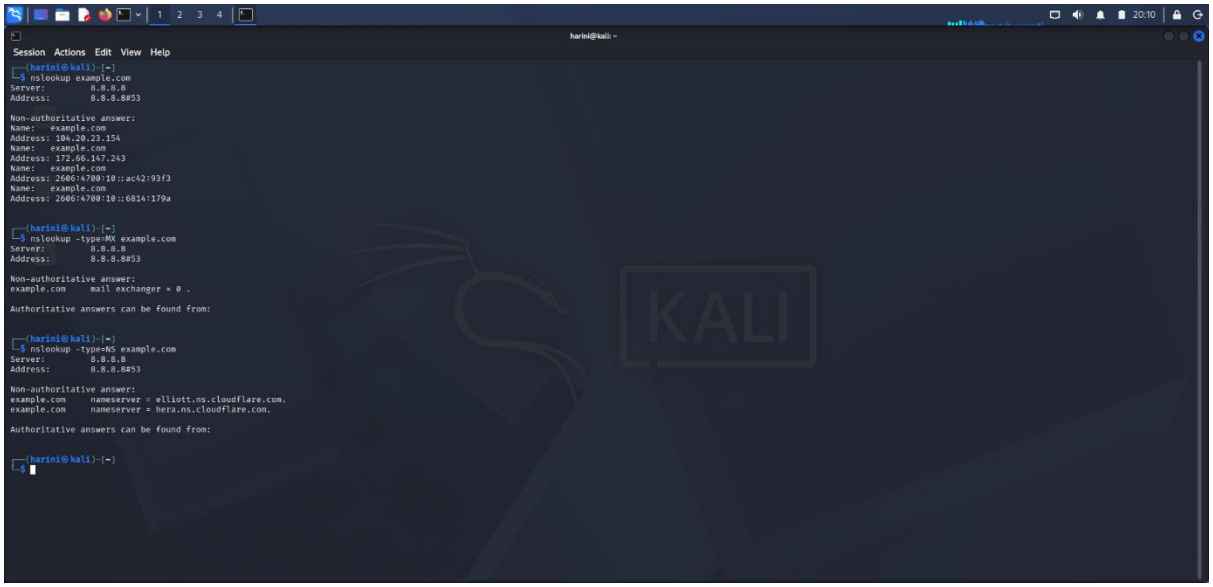
nslookup example.com

nslookup -type=MX example.com

nslookup -type=NS example.com

Observation: The commands executed successfully and provided domain resolution information along with mail exchange (MX) and name server (NS) records.

This activity helped me understand how DNS queries can reveal information about a domain's network configuration and supporting infrastructure.



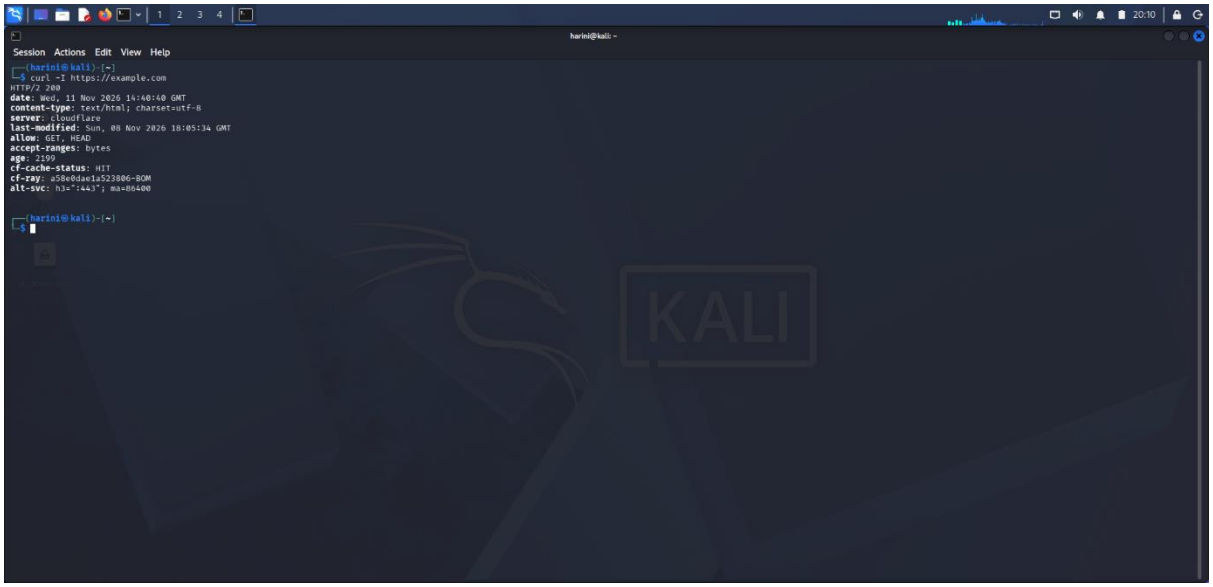
4. cURL – HTTP Response Header Analysis

Objective: Retrieve and examine the HTTP response headers returned by the target website.

Command used: curl -I https://example.com

Observation: The cURL command was used to send an HTTP HEAD request to the target website and retrieve its response headers without downloading the complete webpage. The response headers provide information about the server's HTTP response, content type, caching behaviour, and other web-related details, depending on the headers returned.

This activity helped me understand how cURL can be used to inspect HTTP responses and gather technical information during web reconnaissance.



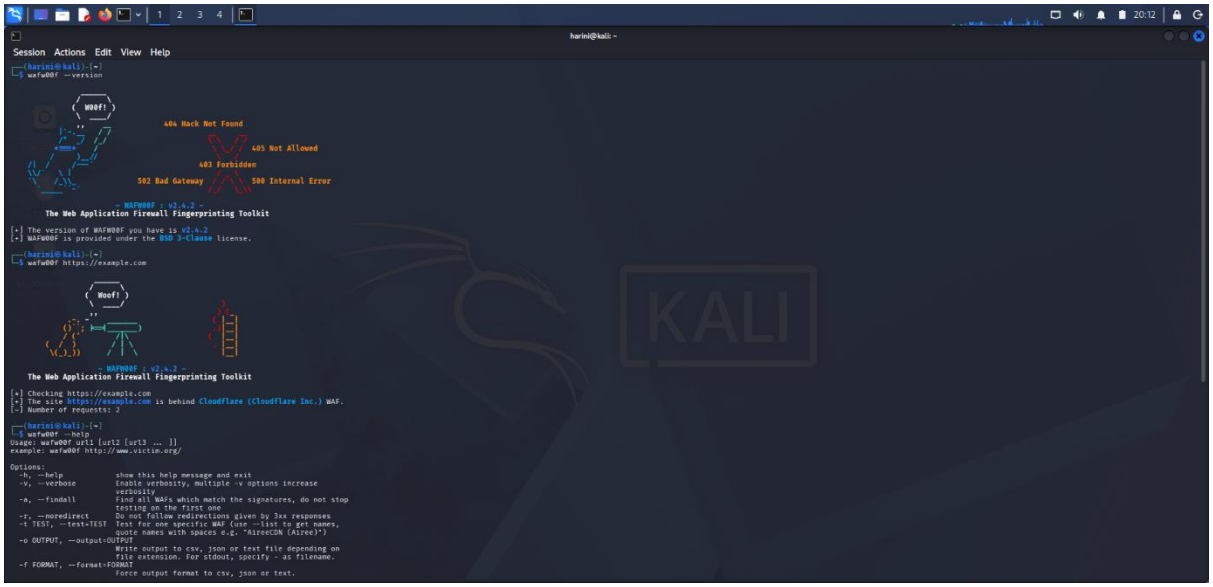
5. Wafw00f – Web Application Firewall Detection

Objective: Identify whether a Web Application Firewall was protecting the target website.

Command used: wafw00f https://example.com

Observation: The tool executed successfully using Wafw00f version 2.4.2. The result identified **Cloudflare (Cloudflare Inc.)** as the Web Application Firewall protecting the website.

This activity demonstrated how WAF detection tools can help identify security mechanisms deployed in front of web applications.



6. DNSRecon – DNS Enumeration

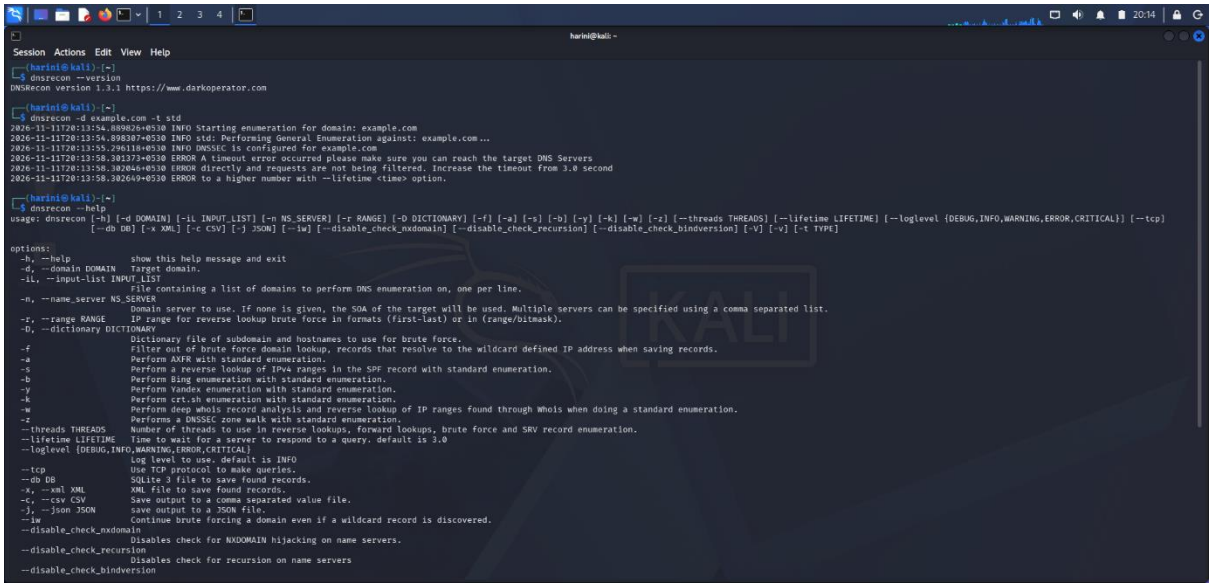
Objective: Perform DNS enumeration against the target domain and collect additional DNS-related information.

Command used: dnsrecon -d example.com -t std

Observation:

The command executed successfully using DNSRecon version 1.3.1 and displayed DNS-related enumeration results.

This activity helped me understand how DNS enumeration can be used to gather information about a domain's infrastructure and associated DNS records.



4.2 W2-PM5: Network Scanning with Zenmap

For the second activity, I used **Zenmap, the graphical interface for Nmap, on Windows** to perform network scanning and host discovery against my own local system. The practical involved exploring the Zenmap interface, checking host availability, examining TCP ports, performing service version detection, and attempting operating system detection. I also examined the Windows IP configuration and network adapter MAC addresses.

1. Zenmap Installation and Interface Exploration

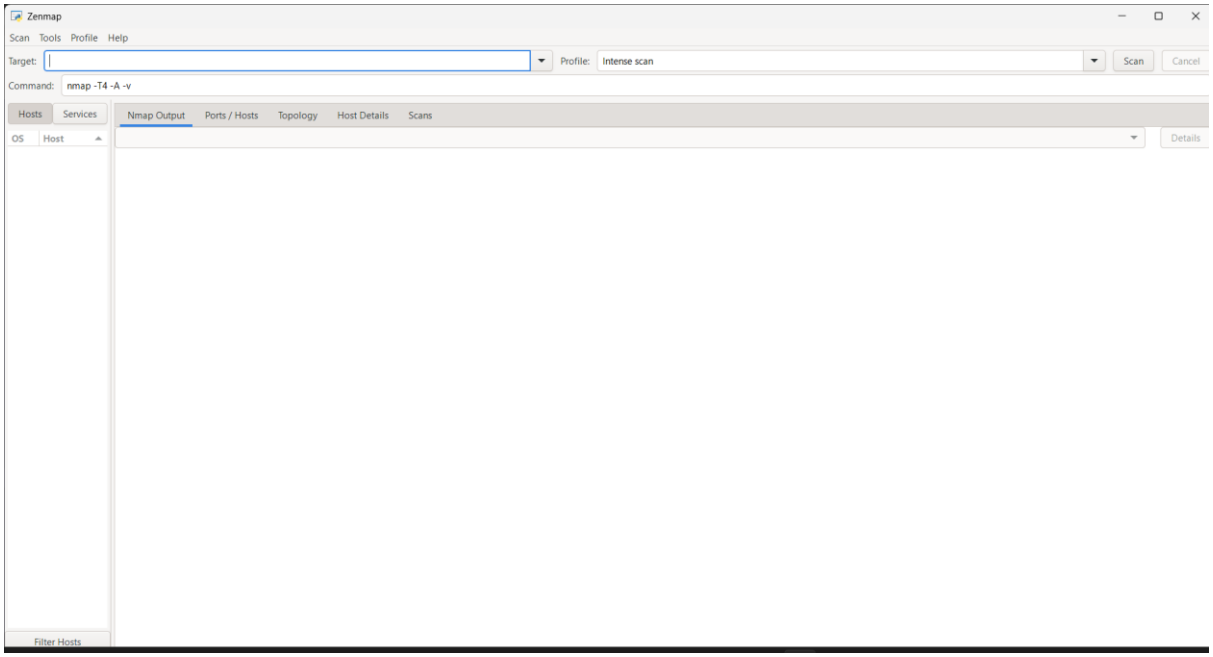
Objective: Install Zenmap and explore its graphical interface and available scanning profiles.

Key Activities:

- Installed and opened Zenmap on Windows.
- Explored the Target field and Profile selection.
- Examined the scan command and output sections.
- Explored available profiles, including Ping Scan, Regular Scan, and Quick Scan.

Observation:

The Zenmap interface provided a graphical way to configure Nmap scans, execute commands, and examine scan results.



2. Windows IP Configuration

Objective: Examine the local system's network configuration.

Command used: ipconfig

Key Activities:

- Examined available network interfaces.
- Identified the active Wi-Fi IPv4 configuration.
- Examined the subnet mask and default gateway.
- Identified the VirtualBox Host-Only network adapter.

Observation:

The command displayed the network configuration of the Windows system, helping me understand the relationship between an IP address, subnet mask, gateway, and network interface.

The college Wi-Fi network was not scanned because authorization to scan other connected devices was unavailable.

```
C:\Users\Harini Madasu>ipconfig | findstr /i "IPv4 Subnet Default Gateway"
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   IPv4 Address. . . . . . . . . . . : 172.16.18.84
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.16.16.1
```

3. Ping Scan – Host Discovery

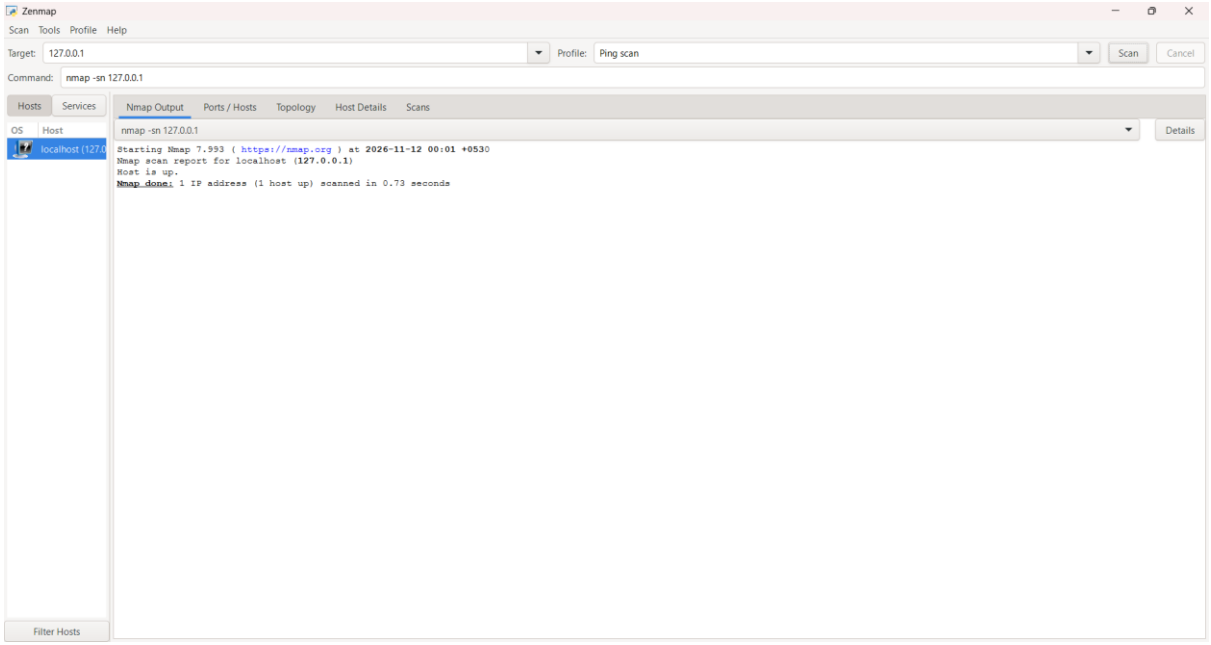
Objective: Check whether the local host was active and reachable.

Target: 127.0.0.1

Command used: nmap -sn 127.0.0.1

Observation: The scan successfully identified the localhost target as up.

This confirmed the availability of the local system through its loopback address. The scan was restricted to localhost and did not identify other devices on the college network.



4. MAC Address Identification

Objective: Identify the MAC addresses associated with the local system's network adapters.

Command used: getmac /v

Key Activities:

- Examined the available network adapters.
- Identified the physical address associated with the Wi-Fi adapter.
- Examined the VirtualBox Host-Only adapter's MAC address.

Observation: The command displayed the MAC addresses associated with the system's network interfaces.

These addresses represent my own network adapters and are not MAC addresses discovered from other LAN devices.

```
C:\Users\Harini Madasu>getmac /v

Connection Name  Network Adapter  Physical Address    Transport Name
===============  ===============  ==================  ============================================================
Wi-Fi            Realtek 8851BE   94-BB-43-83-E8-A8   \Device\Tcpip_{86565F10-52EE-476E-BB89-64C21AB83D58}
Ethernet         Realtek PCIe Gb  D4-93-90-40-66-9E   Media disconnected
Ethernet 2       VirtualBox Host  0A-00-27-00-00-10   \Device\Tcpip_{D1A7270C-8C58-47EB-A6AE-0E18565DE930}
```

5. Service Version Detection

Objective: Identify services and their version information on open TCP ports.

Command used: nmap -sV 127.0.0.1

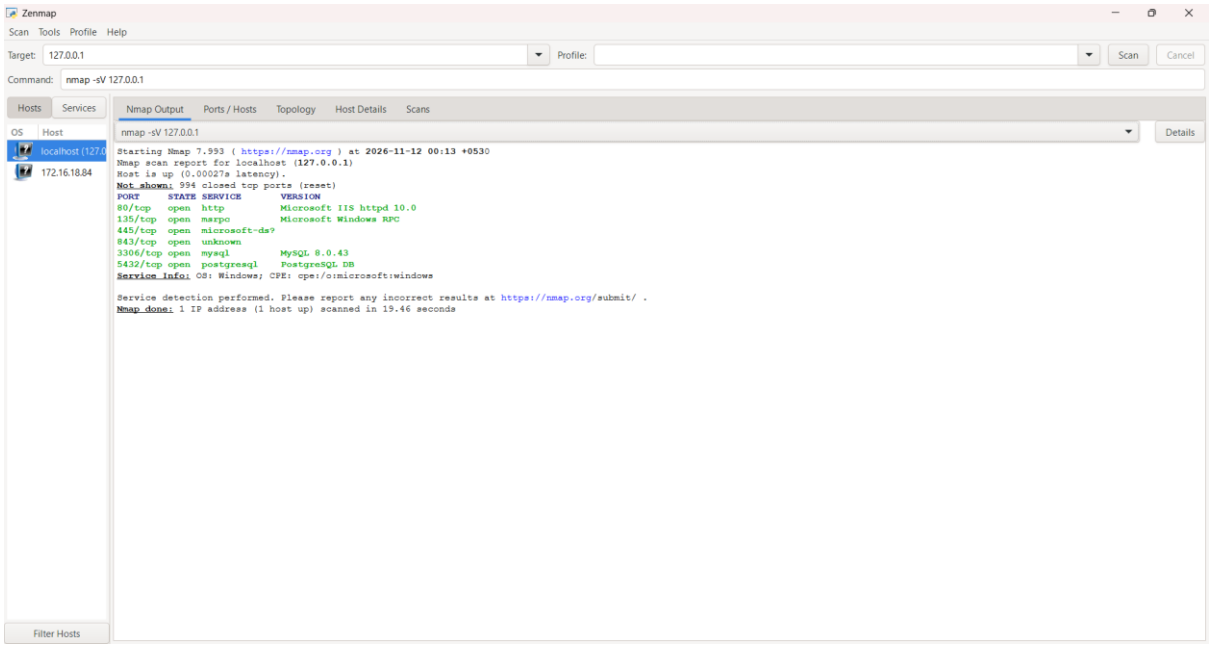
Observation: The scan identified the localhost system as up and reported six open TCP ports and 994 closed TCP ports among the default 1,000 scanned TCP ports.

The following services were identified:

Port	State	Service	Detected Information
80/tcp	Open	HTTP	Microsoft IIS httpd 10.0
135/tcp	Open	MSRPC	Microsoft Windows RPC
445/tcp	Open	Microsoft-DS	Version not identified
843/tcp	Open	Unknown	Service not identified
3306/tcp	Open	MySQL	MySQL 8.0.43
5432/tcp	Open	PostgreSQL	PostgreSQL DB

The scan also provided service information indicating a Windows operating system.

This activity demonstrated how Nmap's -sV option can identify services and provide version information about accessible ports.



6. Operating System Detection

Objective: Explore operating system fingerprinting using Nmap.

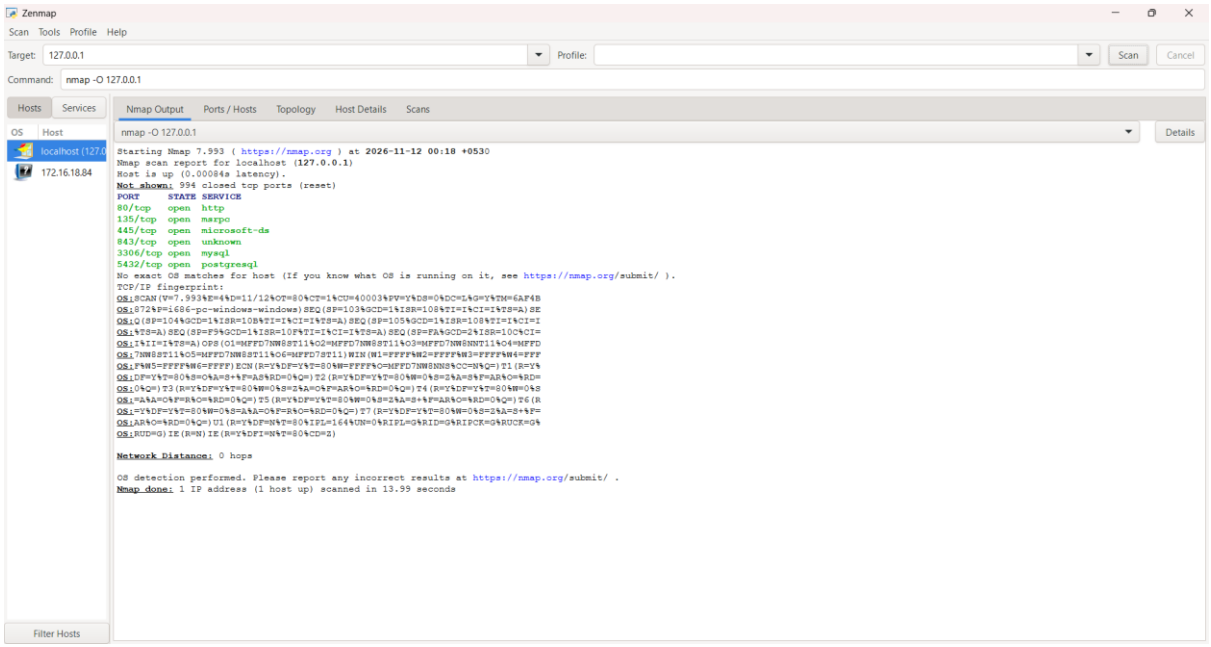
Command used: nmap -O 127.0.0.1

Key Activities:

- Used the -O option to enable operating system detection.
- Examined the host status and TCP port information.
- Explored Nmap's operating system fingerprinting capability.

Observation: The scan was performed successfully. However, a localhost scan may not provide a reliable operating system fingerprint. Therefore, the result was not treated as a definitive OS identification.

This activity helped me understand the purpose and limitations of operating system detection during network scanning.



7. Saving Nmap Scan Results

Objective: Preserve scan output for documentation and further analysis.

Scan command: nmap -sV 127.0.0.1

Output File: zenmap_local_scan.txt

Key Activities:

- Executed the service version detection scan.
- Preserved the scan output in a text file.

- Reviewed the saved results for documentation and future reference.

Observation: The scan results were saved as zenmap_local_scan.txt on the Windows Desktop for further reference.

8. Network Topology Visualization

Objective: Examine the topology visualization available in Zenmap.

Key Activities:

- Opened the Topology tab in Zenmap.
- Examined the visualization generated from the localhost scan.
- Captured the topology view and saved it as a PDF.

Output File: Network_Topology.pdf

Observation:

The topology view represented the localhost scan.

---

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, the following observations and their potential security implications were identified.

No.	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Domain registration information available	WHOIS retrieved publicly available domain information.	May help an attacker understand domain registration and infrastructure details.	Low
2	Website technology information identifiable	WhatWeb successfully fingerprinted the target website.	Technology information may help attackers identify software requiring further security review.	Low
3	DNS information publicly available	Nslookup retrieved domain resolution, MX, and NS information.	DNS information may assist in mapping domain infrastructure.	Low

4	WAF technology identifiable	Wafw00f identified Cloudflare as the WAF.	Reveals information about the website's protective infrastructure.	Low
5	DNS infrastructure information discoverable	DNSRecon successfully performed DNS enumeration.	DNS records may provide information useful for further reconnaissance.	Low
6	Localhost reachable	Nmap Ping Scan reported 127.0.0.1 as up.	Confirms local host availability but does not indicate external network exposure.	Informational
7	Open TCP ports identified	Service detection identified six open TCP ports on localhost.	The services may require configuration and security review.	Informational
8	OS fingerprinting limitations	The localhost OS detection exercise did not establish a definitive fingerprint.	Additional suitable scan conditions may be required for reliable OS identification.	Informational

Risk Level Key: Critical | High | Medium | Low | Informational

The observations above are based on the results of the assigned footprinting and scanning exercises. They do not represent confirmed vulnerabilities. The identification of publicly available information, a WAF, or open local ports does not automatically indicate a security weakness. Further authorized testing would be required to establish whether any actual vulnerability exists.

No exploitation, vulnerability validation, or unauthorized access to external systems was performed.

6. Recommendations

Based on the observations from the footprinting and network scanning activities, the following security recommendations can be considered:

1. Review Publicly Available Domain Information

Organizations should periodically review domain registration information and technical details that are publicly accessible.

2. Keep Website Technologies Updated

Web servers, CMS platforms, plugins, and other software components should be updated regularly to reduce exposure to known vulnerabilities.

3. Monitor DNS Configuration

DNS records should be reviewed periodically to ensure that only required records and services are publicly exposed.

4. Maintain Proper WAF Configuration

Web Application Firewalls should be configured, maintained, and monitored to help protect web applications against malicious traffic.

5. Perform Regular Authorized Network Scanning

Organizations should conduct authorized network scans periodically to identify active hosts, open ports, and potentially unnecessary services.

6. Review Open Network Ports

Open ports should be reviewed to ensure that the associated services are necessary and properly secured.

7. Investigate Unexpected Services

Unknown or unexpected services discovered during network scanning should be investigated and verified by the responsible system administrator.

8. Maintain Scan Documentation

Network scan results should be saved and documented to support future security reviews and comparisons.

9. Follow Authorized Testing Practices

Footprinting, reconnaissance, and network scanning should always be conducted within an approved scope and with appropriate authorization.

7. Conclusion

During Week 2 of my cybersecurity training at Network Walks, I completed practical activities covering web footprinting, reconnaissance, and network scanning.

In the footprinting activity (W2-PM1), I used WHOIS, WhatWeb, Nslookup, Wafw00f, and DNSRecon to collect publicly available domain, website, and DNS-related information about example.com. I learned how these tools can be used to understand domain registration details, identify website technologies, retrieve DNS records, perform DNS enumeration, and detect Web Application Firewalls.

In the network scanning activity (W2-PM5), I used Zenmap and Nmap on Windows to perform host discovery and scanning against my own local system using the address 127.0.0.1. I explored the Zenmap interface and scanning profiles, performed a Ping Scan, examined TCP ports, identified services and their versions, and attempted operating system detection. I also examined the Windows network configuration and MAC addresses, saved scan results in a text file, and generated a localhost topology PDF.

The network scanning exercise was restricted to my own system. LAN-wide host discovery was not performed because authorization to scan other devices on the college network was unavailable.

These practical exercises helped me understand the initial stages of penetration testing and the importance of reconnaissance and network discovery in cybersecurity. I also gained practical experience in using command-line tools, interpreting scan results, troubleshooting, and documenting technical findings.

Overall, Week 2 provided valuable hands-on experience with Kali Linux, Zenmap, and Nmap. I would like to thank **Network Walks and Wakis Sir** for providing the learning resources, practical guidance, and support throughout the training.

👤 Author

Harini Madasu
Cybersecurity Trainee
B084 – Network Walks

📌 Project Information

- **Program Name:** Cybersecurity Training Program at Network Walks
- **Week:** 02
- **Modules:** W2-PM1 and W2-PM5
- **Repository:** Network Walks – Week 2 Cybersecurity Projects